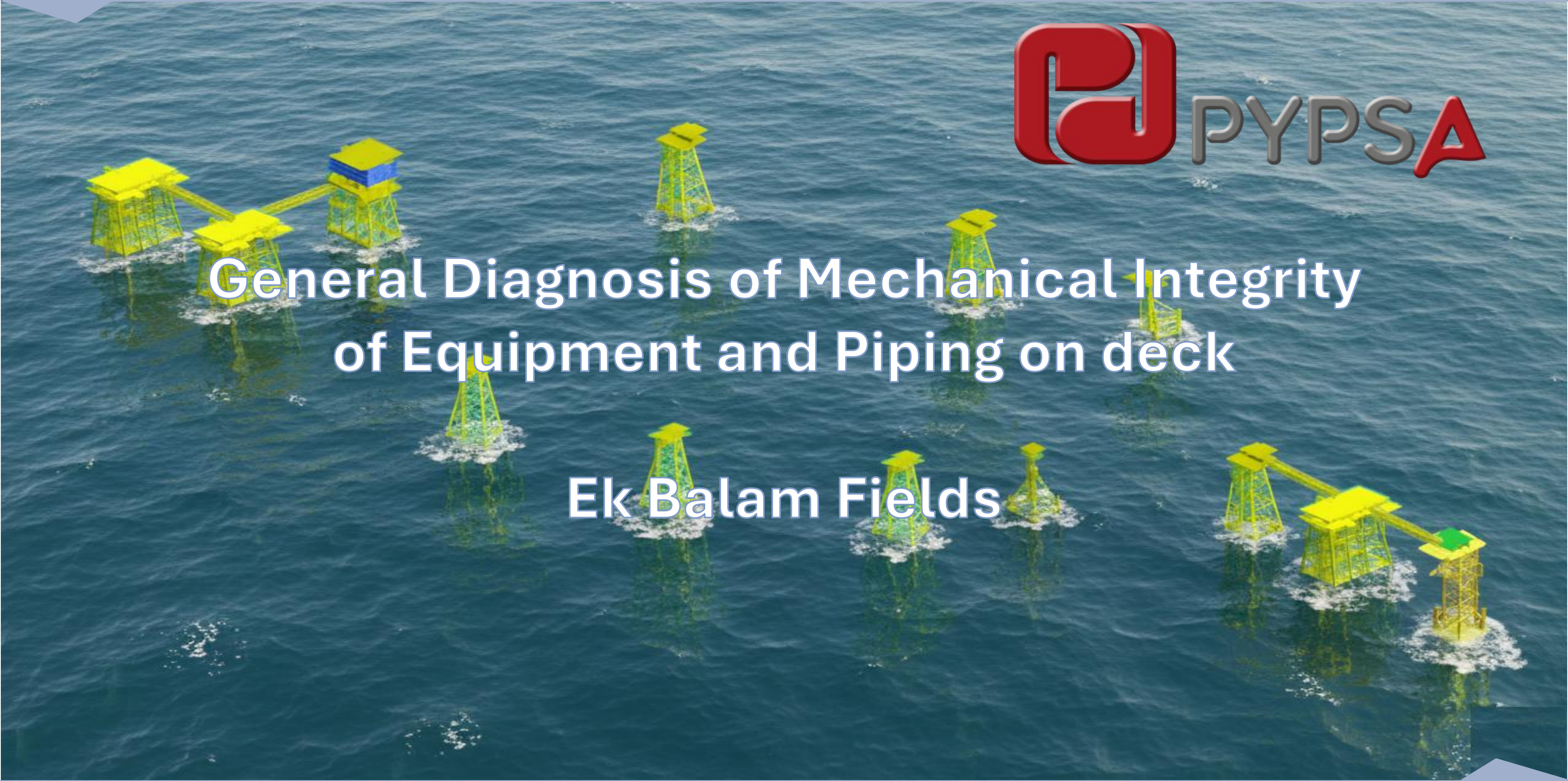


General Diagnosis of Mechanical Integrity of Equipment and Piping on deck

Ek Balam Fields

Location

PYPSA

Scope

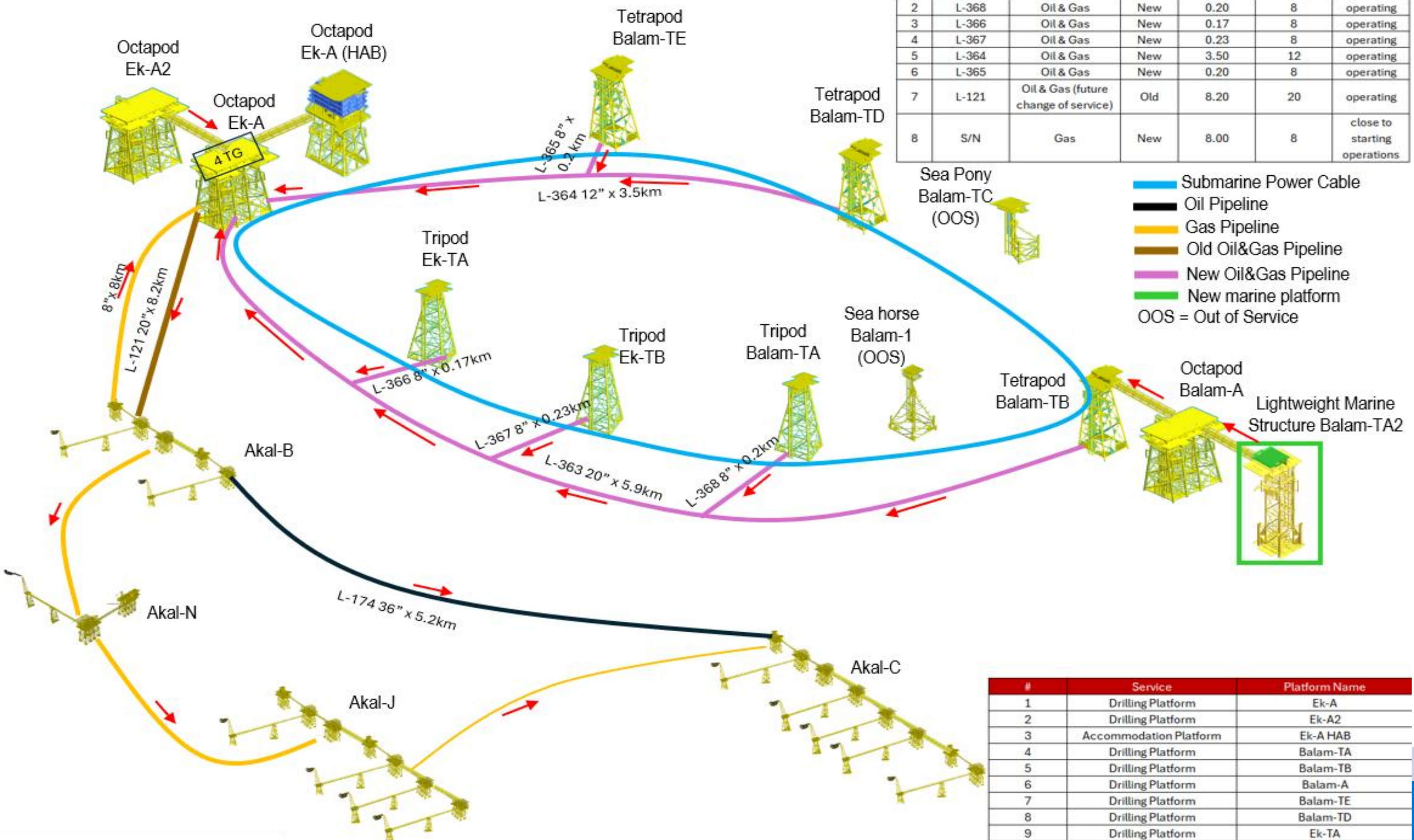
This mechanical integrity analysis covers the offshore platforms listed in the table below, which are part of the Ek Balam field. It presents a general diagnosis of the current condition of the fixed equipment located on deck, including pressure vessels, piping, tanks, filters, level indicators, fin-fan coolers, and heat exchangers.

No.	Offshore Platform	Installation Date	Type	Service	Water Depth (m)	Number of wells	Wells
1	Ek-A	2018	Octapod	Drilling	51	6	Ek-11, Ek-27, Ek-31, Ek-33, Ek-56, Ek-101
2	Balam-TB	2017	Tetrapod	Drilling	47.9	4	Balam-47, Balam-63, Balam-67, Balam-83
3	Balam-A	2014	Octapod	Drilling	47.9	6	Balam-32, Balam-41, Balam-42, Balam-61, Balam-85, Balam-93
4	Ek-A2	2018	Octapod	Drilling	51	7	Ek-14, Ek-29, Ek-32, Ek-47, Ek-53, Ek-57, Ek-69
5	Ek-TA	1996	Tripod	Drilling	49.9	5	Ek-9, Ek-17, Ek-21, Ek-25, Ek-37
6	Balam-TE	2020	Tetrapod	Drilling	51.2	5	Balam-71, Balam-75, Balam-91, Ek-35H, Ek-55
7	Balam-TA	2021	Tripod	Drilling	48	2	Balam-2, Balam-22
8	Balam-TA2	2022	Tetrapod (Lightweight Marine Structure)	Drilling	47.9	2	Balam-5, Balam-87
9	Ek-TB	1996	Tripod	Drilling	49.1	2	Ek-41, Ek-63
10	Balam-TD	2005	Tetrapod	Drilling	50.3	1	Balam-3
11	Ek-A Habitacional	2018	Octapod	Accomodation	51	0	None

Status of Ek Balam Field (July 2026)

PYPSA

#	Line	Service	Age	Lenght (km)	Diameter (in)	Status
1	L-363	Oil & Gas	New	5.90	20	operating
2	L-368	Oil & Gas	New	0.20	8	operating
3	L-366	Oil & Gas	New	0.17	8	operating
4	L-367	Oil & Gas	New	0.23	8	operating
5	L-364	Oil & Gas	New	3.50	12	operating
6	L-365	Oil & Gas	New	0.20	8	operating
7	L-121	Oil & Gas (future change of service)	Old	8.20	20	operating
8	S/N	Gas	New	8.00	8	close to starting operations

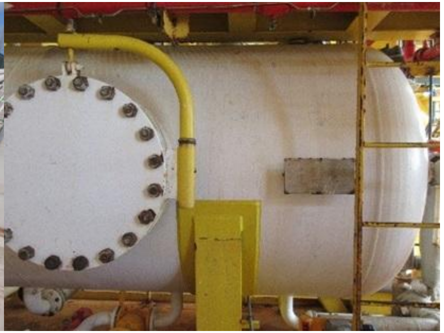


#	Service	Platform Name
1	Drilling Platform	Ek-A
2	Drilling Platform	Ek-A2
3	Accommodation Platform	Ek-A HAB
4	Drilling Platform	Balam-TA
5	Drilling Platform	Balam-TB
6	Drilling Platform	Balam-A
7	Drilling Platform	Balam-TE
8	Drilling Platform	Balam-TD
9	Drilling Platform	Ek-TA
10	Drilling Platform	Ek-TB
11	Drilling Platform	Balam-TA2

Criteria used for the analysis

In this analysis, static equipment is classified into 8 states that qualitatively determine its mechanical integrity; these states are: SC, NC, RSC, RNC, NI, CID, NRLt, and OOS

SC.- Standard Compliant*: The item meets design specifications, exhibits acceptable integrity, shows no corrosion or only minimal acceptable corrosion, has corrosion protection in good condition, adequate and standardized supports, bolts and nuts in good condition, no hydrocarbon stains, valve handles for operation, inspected welds in good condition, and a grounding system. This condition can only be determined through non-destructive testing (NDT) according to ASNT with the approval of an API-certified expert or an ASNT-certified inspector.*



NC.- Non-Compliant: *The item does not meet one or more of the following criteria: design specification, mechanical integrity condition, corrosion protection, supports, condition of studs and nuts, cleanliness, valve handles, weld condition, grounding system. This evaluation just can be give it through NDT according ASNT with API-certified expert or ASNT-certified inspector validation. A nonconformity must be addressed immediately; however, only an API or ASNT inspector can determine the risk level of the NC.*

Criteria used for the analysis

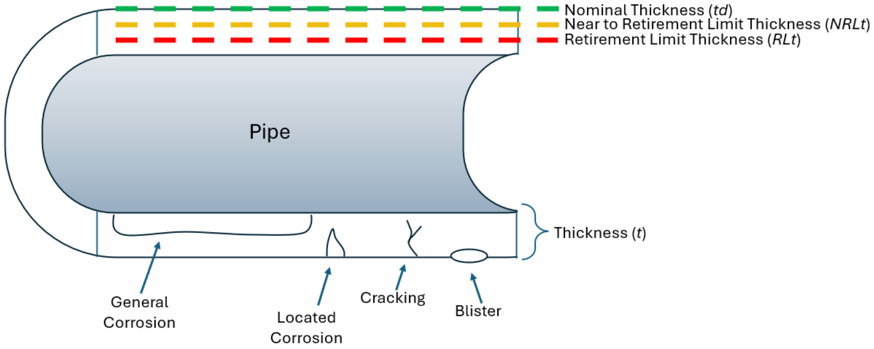
RSC or RNC.- Repaired Standard-Compliant or Repaired Non-Compliant : *This means that the element had a non-compliance (NC), but it was addressed and repaired, bringing the element to a state of SC.*

NI.- Not Inspected: *An item that is not inspected automatically becomes a NC; however, an API-certified expert can keep the item's status at NI based on the item's risk.*

Criteria used for the analysis

***CID.- Corrosion-Induced Degradation**: It refers to the process where chemical or electrochemical reactions (corrosion) break down a material, directly resulting in a loss of its physical, structural, or load-bearing properties. This condition already represents a high risk which must be addressed immediately considering the risk it represents.*

***NRLt.- Near to Retirement Limit thickness**: this is thickness near to minimum required for an in-service component (pipe, vessel, or tank) to safely operate at its design pressure and temperature.*



***OOS.- Out Of Service**: This refers to the item that is not in operation and does not represent a risk of leakage or loss of containment in the facility.*

PYPSA

The eight mechanical integrity status (SC, NC, RSC, RNC, NI, CID, NRLt, and OOS) were qualitatively grouped into five understandable criteria to determine the percentage of elements in Good, Fair, Near Poor, Poor, and Out of Operation condition. Piping and equipment with a Good or Fair integrity level are considered acceptable, while those with a Near Poor or Poor integrity level are unacceptable. Equipment and piping with the latter two levels must be addressed promptly to prevent leaks and/or loss of containment, as these can cause damage to personnel, facilities, production, and/or the environment, thus requiring financial investment for their correction. Finally, equipment and piping with an "out of operation" level do not represent a leak risk and are therefore considered acceptable. These criteria are represented as follows:

SC	Standard Compliant
RSC	Repaired Standard-Compliant
RNC	Repaired Non-Compliant
NI	Not inspected
CID	Corrosion-Induced Degradation
NRLt	Near to Retirement Limit thickness
NC	Non Compliant
OOS	Out of Operation

Good	Acceptable
Fair	Acceptable
Near Poor	Not Acceptable
Poor	Not Acceptable
OOS	Acceptable

Note: The status Not Inspected (NI) should be considered Non-Conforming; however, the specific criteria used in this analysis determine that the uninspected lines and equipment contain non-hazardous fluids due to the nature of the service they serve. For this reason, it is considered Acceptable.

01.- Ek-A *Type of Structure:* Octapod / *Year of Installation:* 2018 / *Number of Wells:* 6

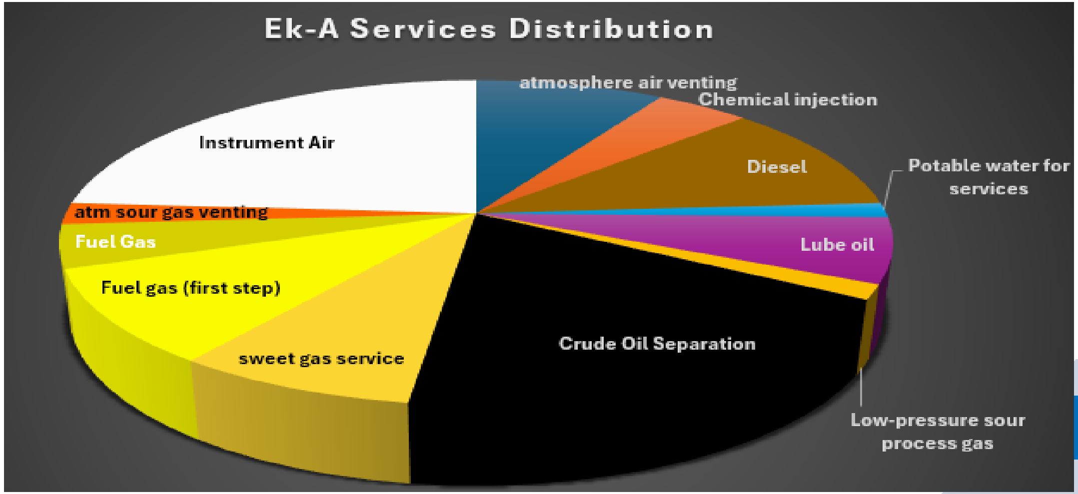
Side view

Bird's eye view

Number of items and their distribution across different services

ITEM \ SERVICE	Atm air venting	Chemical injection	Diesel	Potable water for services	Lube oil	Low-pressure sour process gas	Crude Oil Separation	sweet gas service	Fuel gas (first step)	Fuel Gas	Atm sour gas venting	Instrument Air	Subtotal
Piping circuit	12	4	9	2	6	2	24	8	5	4	3	24	**103**
Chemical tank	0	2	0	0	0	0	0	0	0	0	0	0	**2**
Diesel Filter	0	0	1	0	0	0	0	0	0	0	0	0	**1**
Level Gauge	0	0	2	0	0	0	1	1	3	0	0	0	**7**
Diesel tank	0	0	3	0	0	0	0	0	0	0	0	0	**3**
Fin-fan cooler	0	0	0	0	3	0	0	0	0	0	0	0	**3**
oil separator	0	0	0	0	0	0	1	0	0	0	0	0	**1**
gas scrubber	0	0	0	0	0	0	0	2	0	0	0	0	**2**
fuel gas filter	0	0	0	0	0	0	0	0	2	2	0	0	**4**
fuel gas separator	0	0	0	0	0	0	0	0	1	0	0	0	**1**
heater	0	0	0	0	0	0	0	0	2	0	0	0	**2**
air receiver	0	0	0	0	0	0	0	0	0	0	0	5	**5**
air filter	0	0	0	0	0	0	0	0	0	0	0	4	**4**

Total 138



01.- Ek-A **Type of Structure:** Octapod / **Year of Installation:** 2018 / **Number of Wells:** 6

Side view

Bird's eye view

NDT results for each item

STATUS \ SERVICE	atmosphere air venting	Chemical injection	Diesel	Potable water for services	Lube oil	Low-pressure sour process gas	Crude Oil Separation	sweet gas service	Fuel gas (first step)	Fuel Gas	atm sour gas venting	Instrument Air	Subtotal
SC			4			1	11	4	10	2	1	27	**60**
RSC									2				**2**
RNC			2			1	8	4	1	4	1	6	**27**
NI	12	6	9	2	9								**38**
CID							1						**1**
NRLt							1						**1**
NC							3	2			1		**6**
OOS							2	1					**3**

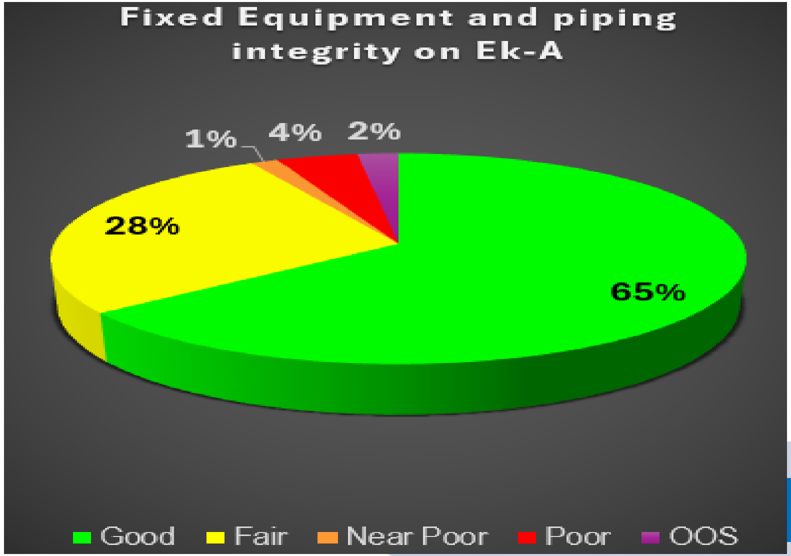
Total 138

NDT Summary

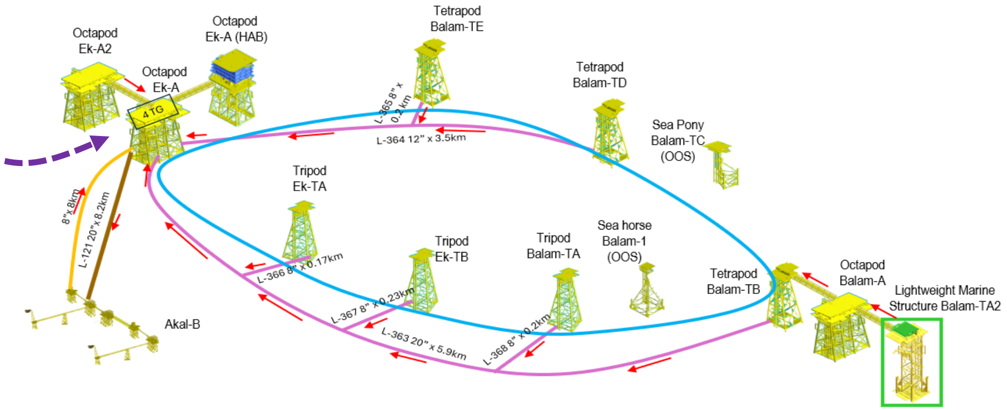
Standard Compliant	60
Repaired Standard-Compliant	2
Repaired Non-Compliant	27
Not inspected	38
Corrosion-Induced Degradation	1
Near to Retirement Limit thickness	1
Non Compliant	6
OOS	3

Integrity Level

Good	89
Fair	38
Near Poor	2
Poor	6
OOS	3
Total	**138**

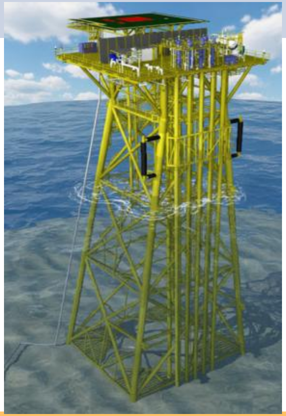


In order of priority, Ek-A is the most strategic platform in the Ek-Balam fields. It operates four Turbogenerators (TGs) that supply electrical power to all platforms. Additionally, it receives all regional production for forwarding to Akal-B and handles the fuel gas supply required for TG operations



Six elements have a non-compliance status; however, two are at imminent risk as they belong to the Sweet Gas Service system. Two pressure vessels (EKA-GS-FA-1112C and EKA-GS-FA-1112A) have maximum operating pressures that exceed their design limits. It is therefore recommended to reduce the maximum operating pressure, install non-metallic reinforcement, or replace the units. Additionally, the vessels exhibit deteriorated anti-corrosion protection and signs of light active corrosion. Historically, combustible gas leaks have had devastating effects on oil facilities

02.- BALAM-TB *Type of Structure:* Tetrapod / *Year of Installation:* 2017 / *Number of Wells:* 4



Side view

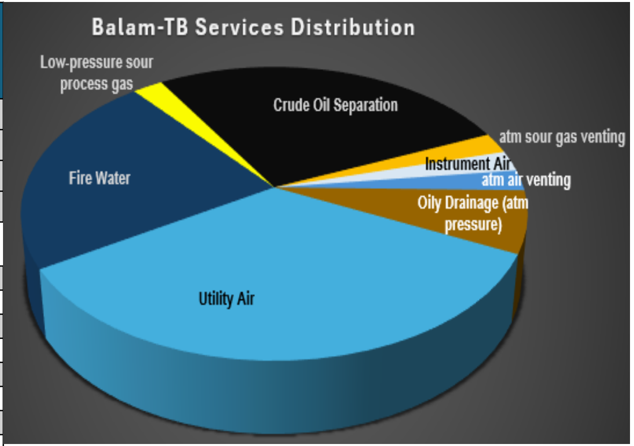
Number of items and their distribution across different services

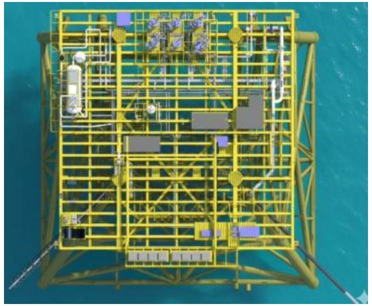
ITEM \ SERVICE	atm air venting	Oily Drainage (atm pressure)	Utility Air	Fire Water	Low-pressure sour process gas	Crude Oil Separation	atm sour gas venting	Instrument Air	Subtotal
Piping Circuit	1	2	10	10	1	11	1	0	**36**
Oily Drainage Tank	0	1	0	0	0	0	0	0	**1**
Air Receiver	0	0	5	0	0	0	0	1	**6**
Oil Separator	0	0	0	0	0	1	0	0	**1**

Total 44

NDT results for each item

	atm air venting	Oily Drainage (atm pressure)	Utility Air	Fire Water	Low-pressure sour process gas	Crude Oil Separation	atm sour gas venting	Instrument Air	Subtotal
SC			4		1	7	1	1	**14**
RSC									**0**
RNC			1			2			**3**
NI	1	3	8	10					**22**
CID									**0**
NRLt									**0**
NC			2			3			**5**
OOS									**0**





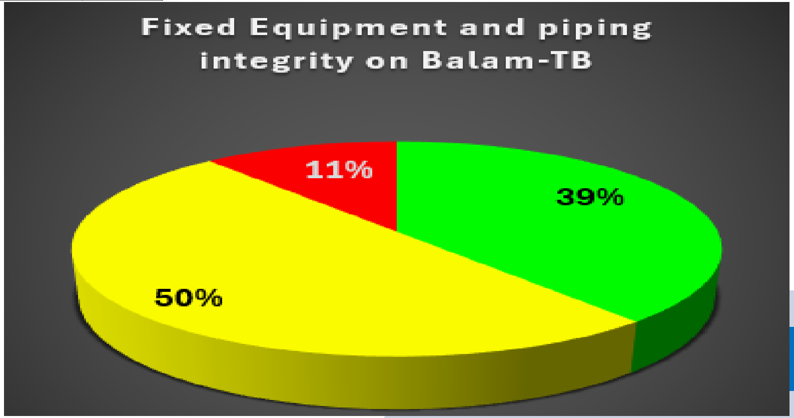
Bird's eye view

NDT Summary

Standard Compliant	14
Repaired Standard-Compliant	0
Repaired Non-Compliant	3
Not inspected	22
Corrosion-Induced Degradation	0
Near to Retirement Limit thickness	0
Non Compliant	5
OOS	0

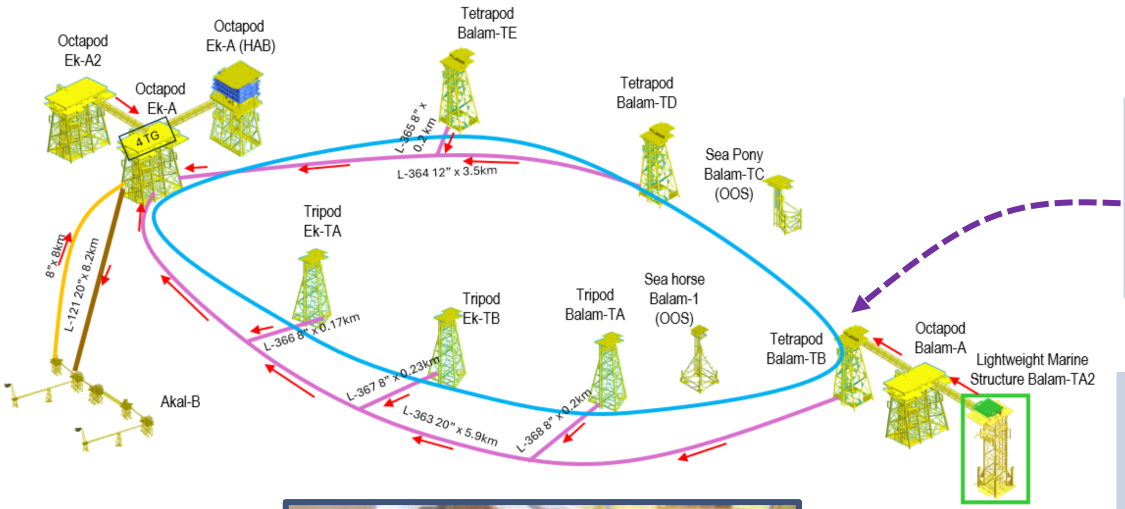
Integrity Level

Good	17
Fair	22
Near Poor	0
Poor	5
OOS	0
Total	**44**



02.- BALAM-TB ***Type of Structure:*** *Tetrapod* / ***Year of Installation:*** *2017* / ***Number of Wells:*** *4*

PYPSA



Balam-TB is the second strategic platform in the Ek-Balam fields because the production from Balam-TA2 and Balam-A passes through it. It is the route for the production of 12 wells in total.

Five elements have a non-compliance status; however, one of them is the most critical. It is a 14" pipe belonging to the crude oil separation system, and its maximum operating pressure is 23 kg/cm² ; this piping circuit has150# flanges that, according to operating conditions, can withstand only up to 19 kg/cm². The solution is to upgrade to Class 300 flanges or reduce the maximum operating pressure. Additionally, severe corrosion is observed on the studs and nuts.

03.- Balam-A *Type of Structure:* Octapod / *Year of Installation:* 2014 / *Number of Wells:* 6

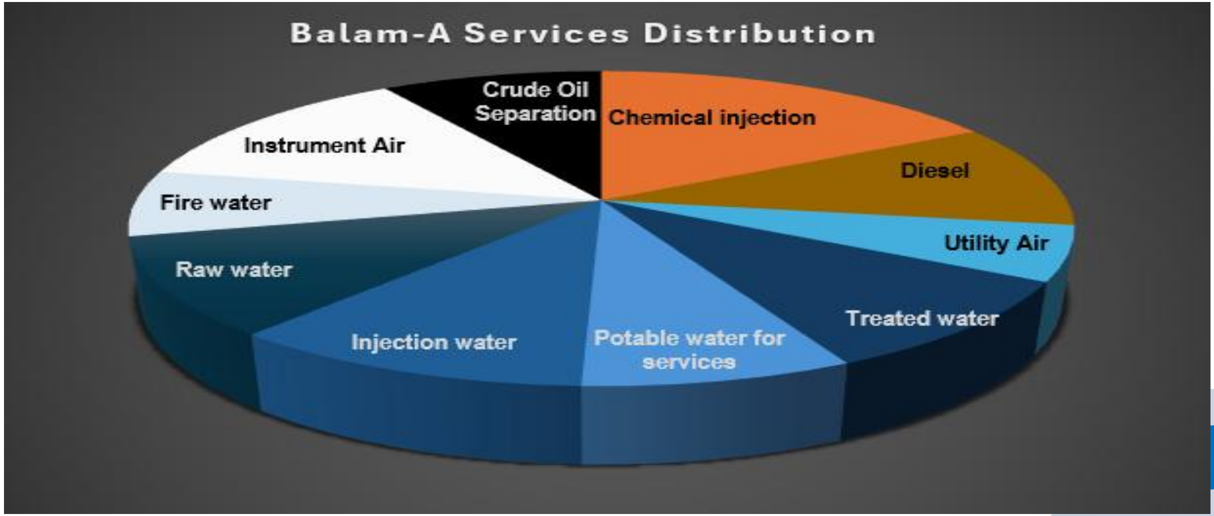
Side view

Bird's eye view

Number of items and their distribution across different services

ITEM \ SERVICE	Chemical injection	Diesel	Utility Air	Treated water	Potable water for services	Injection water	Raw water	Fire water	Instrument Air	Crude Oil Separation	
Level Gauge	5	2	0	0	1	0	0	0	0	0	**8**
Chemical tank	6	0	0	0	0	0	0	0	0	0	**6**
Piping circuit	19	12	9	12	10	19	13	12	22	16	**144**
Diesel tank	0	2	0	0	0	0	0	0	0	0	**2**
Diesel filter	0	3	0	0	0	0	0	0	0	0	**3**
Air Receiver	0	0	1	0	0	0	0	0	1	0	**2**
Water filter	0	0	0	5	1	0	5	0	0	0	**11**
Water tank	0	0	0	0	2	0	0	0	0	0	**2**

Total 178



03.- Balam-A

Type of Structure: *Octapod* / ***Year of Installation:*** *2014* / ***Number of Wells:*** *6*

Side view

Bird's eye view

NDT results for each item

STATUS \ SERVICE	Chemical injection	Diesel	Utility Air	Treated water	Potable water for services	Injection water	Raw water	Fire water	Instrument Air	Crude Oil Separation	Subtotal
SC									1	12	**13**
RSC											**0**
RNC							1		1		**2**
NI	30	19	10	17	14	19	17	12	21		**159**
CID											**0**
NRLt										1	**1**
NC										3	**3**
OOS											**0**

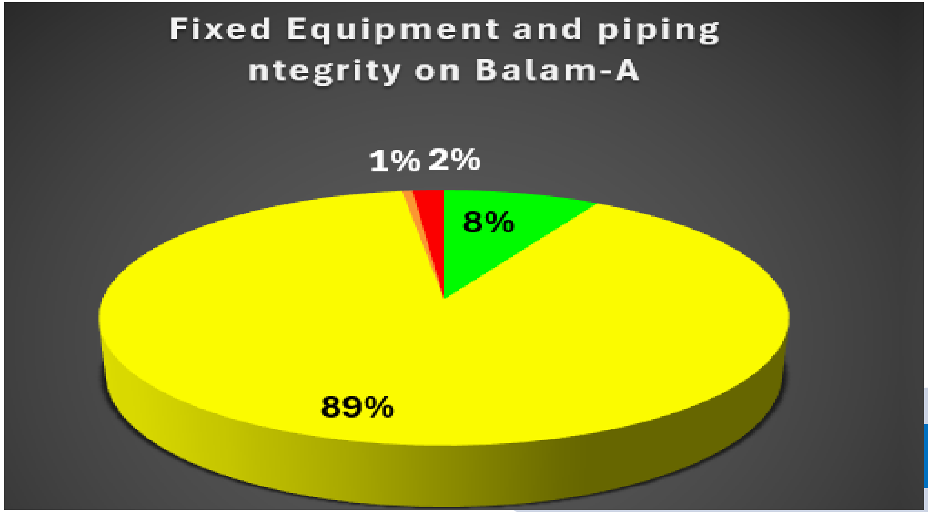
Total 178

NDT Summary

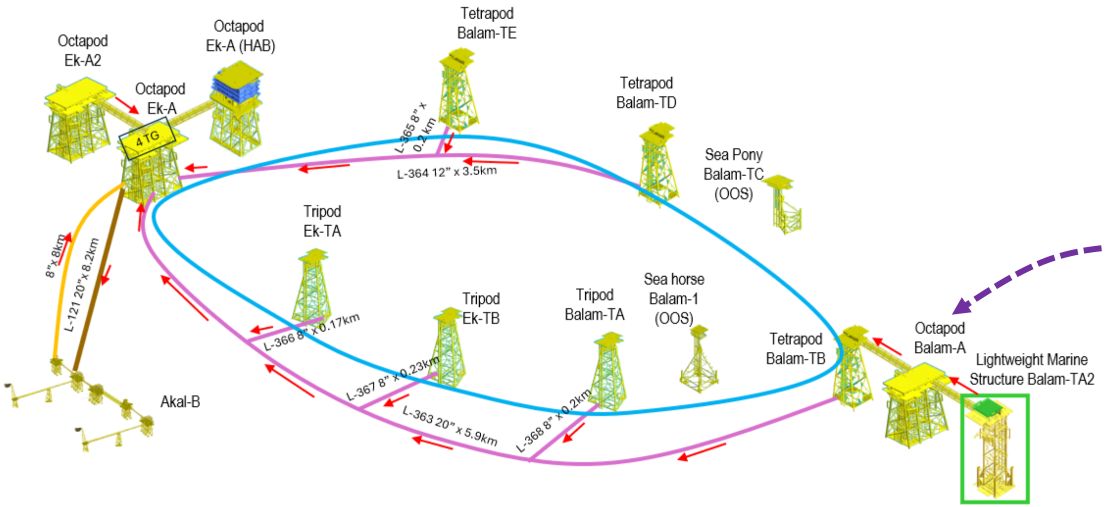
Standard Compliant	13
Repaired Standard-Compliant	0
Repaired Non-Compliant	2
Not inspected	159
Corrosion-Induced Degradation	0
Near to Retirement Limit thickness	1
Non Compliant	3
OOS	0

Integrity Level

Good	15
Fair	159
Near Poor	1
Poor	3
OOS	0
Total	**178**



PYPSA



Balam-A is the next priority; it receives production from Balam-TA2 and, along with its own output, handles a total flow from eight wells to be shipped to Balam-TB

Three elements of the crude oil separation system are in a state of non-compliance and must be repaired: two 6-inch pipes and one 14-inch pipe. The damaged sections must be replaced or reinforced with a non-metallic casing, in addition to surface cleaning and the application of anti-corrosive coating

04.- Ek-A2 *Type of Structure:* Octapod / *Year of Installation:* 2018 / *Number of Wells:* 7

PYPSA

Side view

Bird's eye view

Number of items and their distribution across different services

SERVICE / ITEM	Injection water	Raw water	Fire water	Crude Oil Separation	Subtotal
Piping circuit	14	17	12	17	**60**
Water filter	0	5	0	0	**5**

Total 65

NDT results for each item

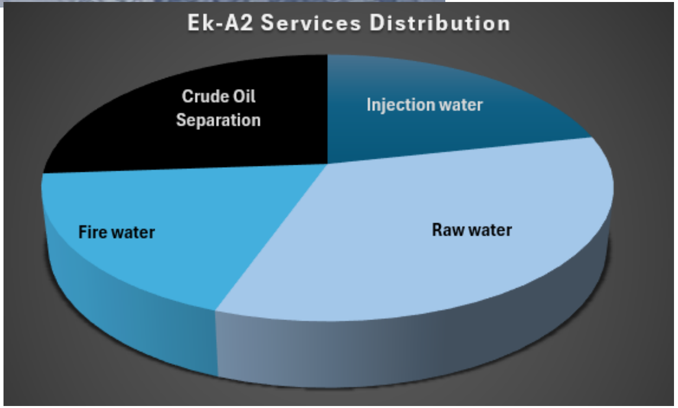
	Injection water	Raw water	Fire water	Crude Oil Separation	Subtotal
SC	9	2		8	**19**
RSC			1		**1**
RNC		2		2	**4**
NI		17	11		**28**
CID	3			6	**9**
NRLt					**0**
NC	2	1		1	**4**
OOS					**0**

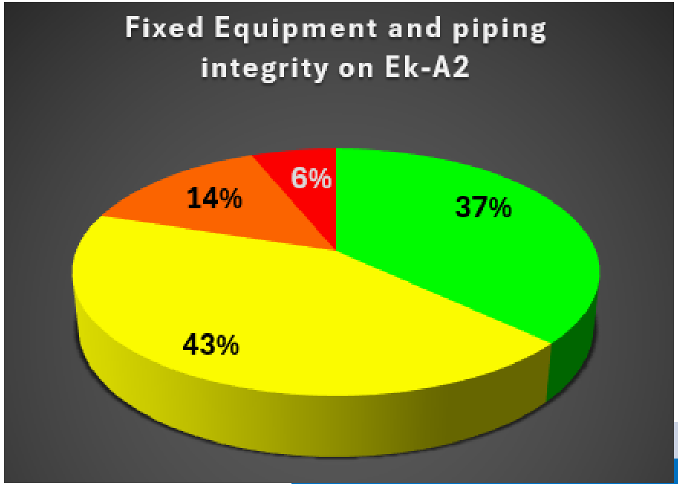
NDT Summary

Standard Compliant	19
Repaired Standard-Compliant	1
Repaired Non-Compliant	4
Not inspected	28
Corrosion-Induced Degradation	9
Near to Retirement Limit thickness	0
Non Compliant	4
OOS	0

Integrity Level

Good	24
Fair	28
Near Poor	9
Poor	4
OOS	0
Total	**65**





05.- Ek-TA **Type of Structure:** *Tripod* / **Year of Installation:** *1996* / **Number of Wells:** *5*

Side view

Bird's eye view

Number of items and their distribution across different services

SERVICE / ITEM	Oily Drainage (atm pressure)	Fire water	Instrument Air	Crude Oil Separation	Subtotal
Oily Drainage Tank	1	0	0	0	**1**
Piping Circuit	2	5	0	18	**25**
Air Receiver	0	0	3	0	**3**

Total 29

NDT results for each item

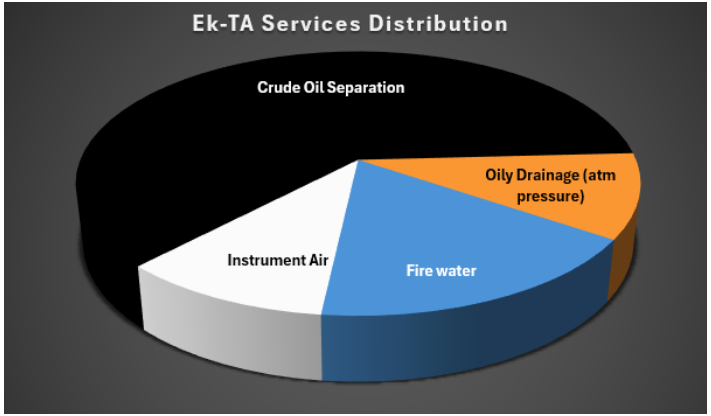
SC			2	13	**15**
RSC					**0**
RNC			1	1	**2**
NI	3	5			**8**
CID					**0**
NRLt				1	**1**
NC				3	**3**
OOS					**0**

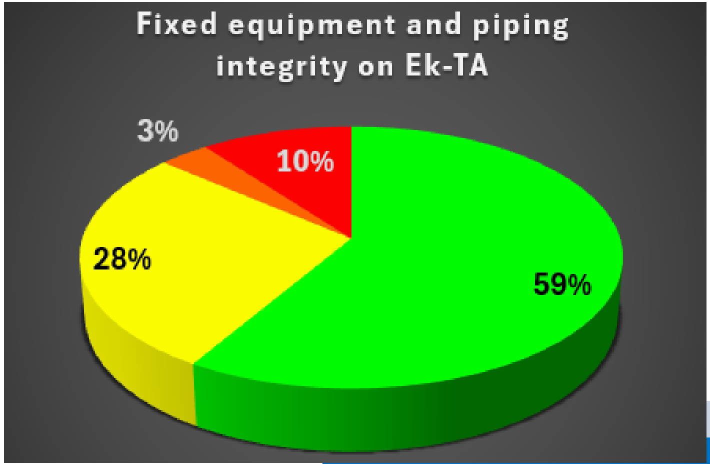
NDT Summary

Standard Compliant	15
Repaired Standard-Compliant	0
Repaired Non-Compliant	2
Not inspected	8
Corrosion-Induced Degradation	0
Near to Retirement Limit thickness	1
Non Compliant	3
OOS	0

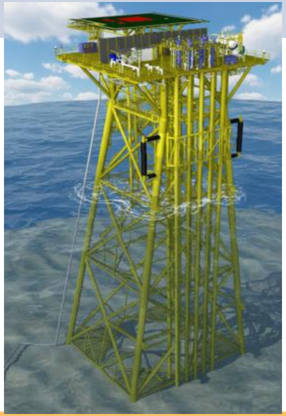
Integrity Level

Good	17
Fair	8
Near Poor	1
Poor	3
OOS	0
Total	**29**





06.- BALAM-TE **Type of Structure:** *Tetrapod* / **Year of Installation:** *2020* / **Number of Wells:** *5*



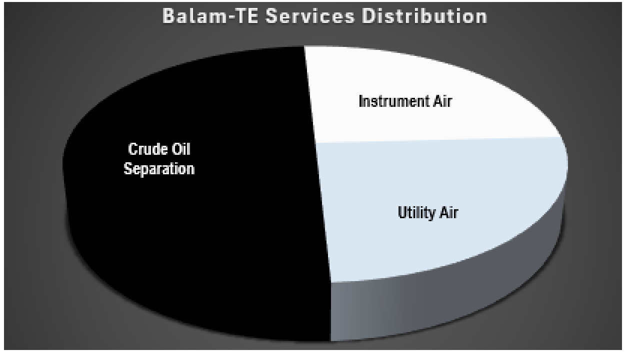
Side view

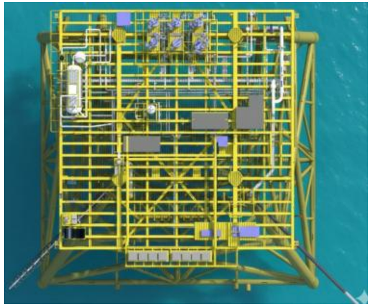
Number of items and their distribution across different services

SERVICE / ITEM	Utility Air	Crude Oil Separation	Instrument Air	Subtotal
Air Receiver	2	0	2	**4**
Piping circuit	2	7	2	**11**
Oil Separator	0	1	0	**1**
				Total 16

NDT results for each item

	Utility Air	Crude Oil Separation	Instrument Air	Subtotal
SC	4	4	3	**11**
RSC				**0**
RNC				**0**
NI				**0**
CID				**0**
NRLt				**0**
NC		3	1	**4**
OOS		1		**1**





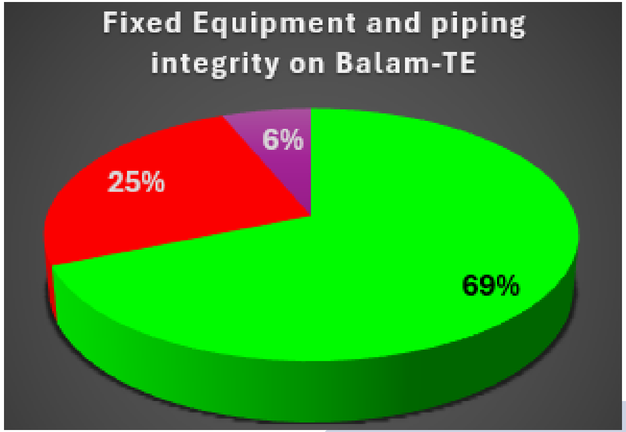
Bird's eye view

NDT Summary

Standard Compliant	11
Repaired Standard-Compliant	0
Repaired Non-Compliant	0
Not inspected	0
Corrosion-Induced Degradation	0
Near to Retirement Limit thickness	0
Non Compliant	4
OOS	1

Integrity Level

Good	11
Fair	0
Near Poor	0
Poor	4
OOS	1
Total	**16**



07.- BALAM-TA *Type of Structure:* Tripod / *Year of Installation:* 2021 / *Number of Wells:* 2

PYPSA

Side view

Bird's eye view

Number of items and their distribution across different services

TYPE \ SERVICE	Utility Air	Crude Oil Separation	Instrument Air	Subtotal
Air Dryer	1	0	0	**1**
Air Receiver	2	0	1	**3**
Piping circuit	2	7	2	**11**
				Total 15

NDT results for each item

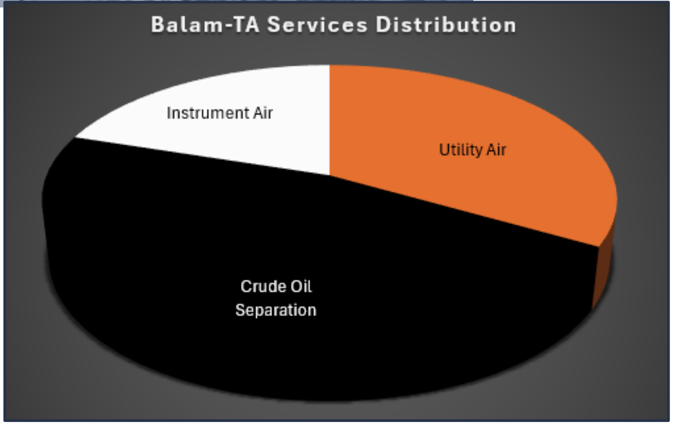
	Utility Air	Crude Oil Separation	Instrument Air	Subtotal
SC	1	4	2	**7**
RSC				**0**
RNC	2		1	**3**
NI	2			**2**
CID		1		**1**
NRLt				**0**
NC		2		**2**
OOS				**0**

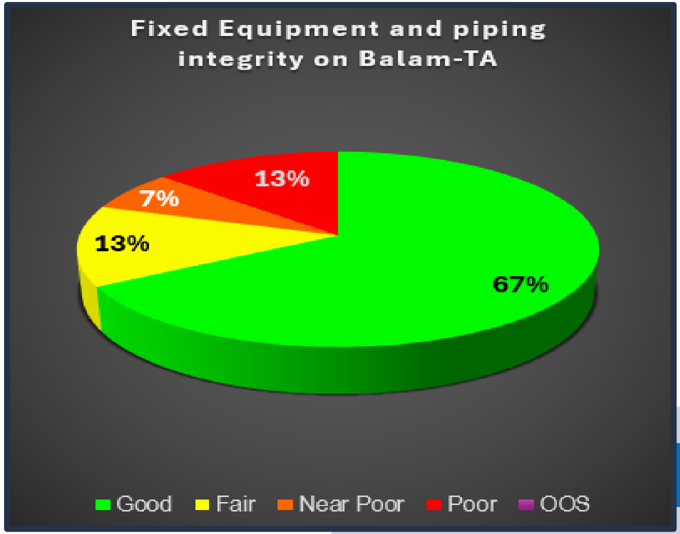
NDT Summary

Standard Compliant	7
Repaired Standard-Compliant	0
Repaired Non-Compliant	3
Not inspected	2
Corrosion-Induced Degradation	1
Near to Retirement Limit thickness	0
Non Compliant	2
OOS	0

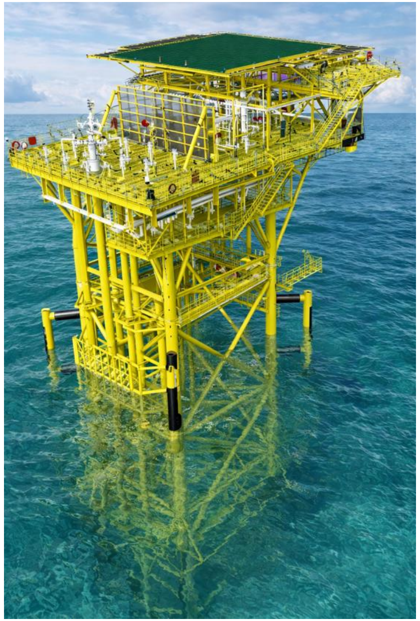
Integrity Level

Good	10
Fair	2
Near Poor	1
Poor	2
OOS	0
Total	**15**

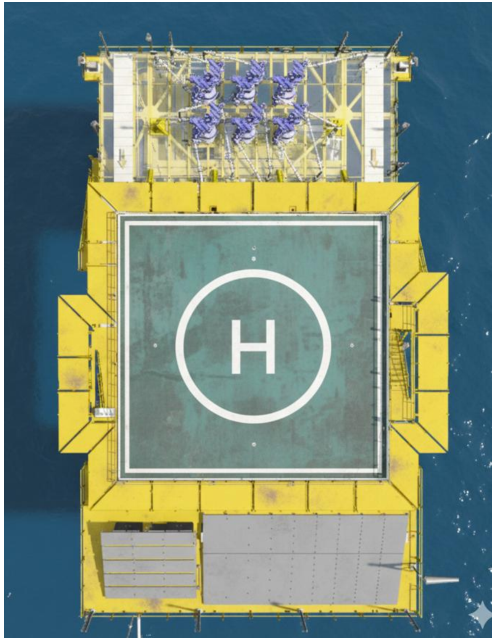




08.- BALAM-TA2 **Type of Structure:** *LMS (Tetrapod)* / **Year of Installation:** *2022* / **Number of Wells:** *2*



Side view



Bird's eye view

This platform was installed in 2022 and has not been inspected because the standard recommends an inspection campaign after three years. Therefore, there is currently no mechanical integrity assessment, and an inspection is required to determine its integrity level. However, the platform is relatively new, although a finding that could put the installation at risk is possible.

09.- Ek-TB **Type of Structure:** *Tripod* / **Year of Installation:** *1996* / **Number of Wells:** *2*

PYPSA

Side view

Bird's eye view

Number of items and their distribution across different services

ITEM \ SERVICE	Chemical injection	Utility Air	Fire water	Crude Oil Separation	Instrument Air	Subtotal
Piping circuit	17	3	2	4	1	**27**
Air filter	0	2	0	0	0	**2**
Air Receiver	0	0	0	0	4	**4**

Total 33

NDT results for each item

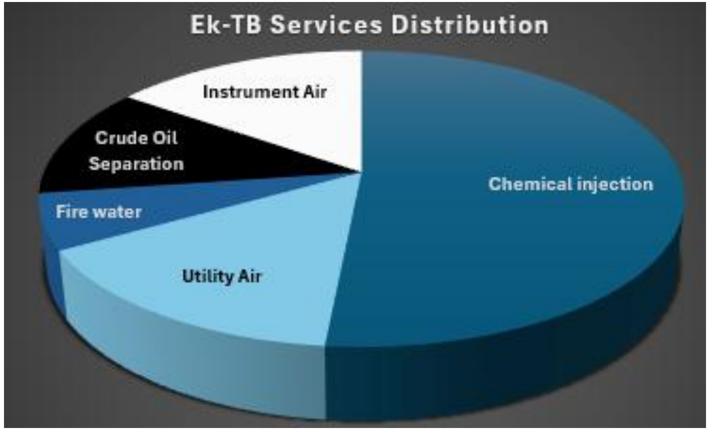
SC		2		1	3	**6**
RSC				1		**1**
RNC						**0**
NI	17	3	2			**22**
CID				1		**1**
NRLt						**0**
NC				1	2	**3**
OOS						**0**

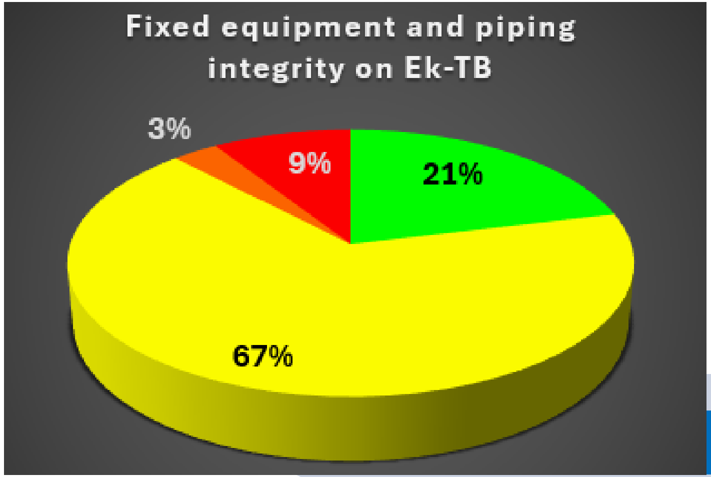
NDT Summary

Standard Compliant	6
Repaired Standard-Compliant	1
Repaired Non-Compliant	0
Not inspected	22
Corrosion-Induced Degradation	1
Near to Retirement Limit thickness	0
Non Compliant	3
OOS	0

Integrity Level

Good	7
Fair	22
Near Poor	1
Poor	3
OOS	0
Total	**33**





10.- BALAM-TD Type of Structure: *Tetrapod* / Year of Installation: *2005* / Number of Wells: *1*

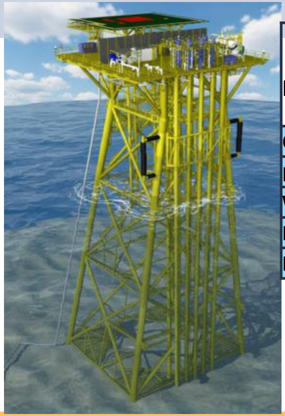
PYPSA

Number of items and their distribution across different services

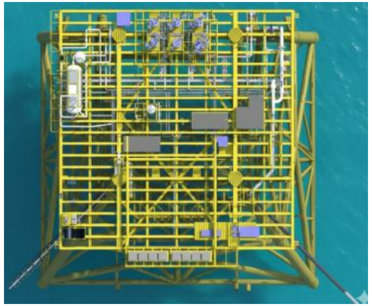
ITEM \ SERVICE	Oily Drainage (atm pressure)	Utility Air	Treated water	Potable water for services	Injection water	Crude Oil Separation	Instrument Air	Subtotal
Oily Drainage Tank	1	0	0	0	1	0	0	**2**
Piping Circuit	4	7	0	4	9	7	3	**34**
Water filter	0	0	1	0	0	0	0	**1**
Level gauge	0	0	0	1	0	0	0	**1**
Hydropneumatic Tank	0	0	0	1	0	0	0	**1**
							Total	**39**

NDT results for each item

	Oily Drainage (atm pressure)	Utility Air	Treated water	Potable water for services	Injection water	Crude Oil Separation	Instrument Air	Subtotal
SC	1	5		1	7	7		**21**
RSC								**0**
RNC		1			3		3	**7**
NI	4	1		5				**10**
CID								**0**
NRLt								**0**
NC			1					**1**
OOS								**0**



Side view



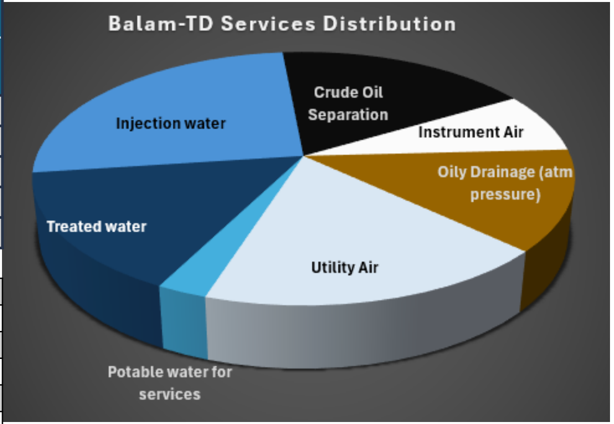
Bird's eye view

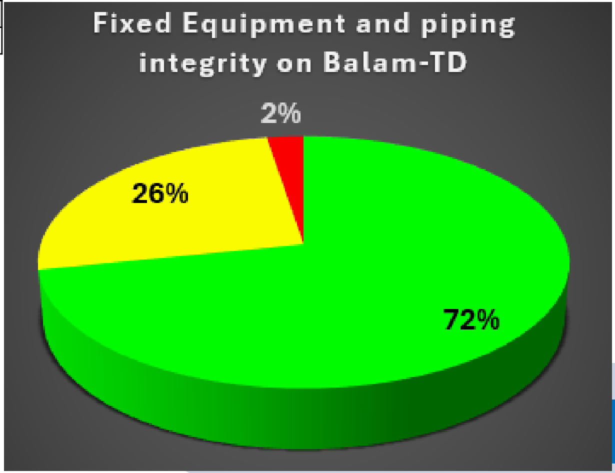
NDT Summary

Standard Compliant	21
Repaired Standard-Compliant	0
Repaired Non-Compliant	7
Not inspected	10
Corrosion-Induced Degradation	0
Near to Retirement Limit thickness	0
Non Compliant	1
OOS	0

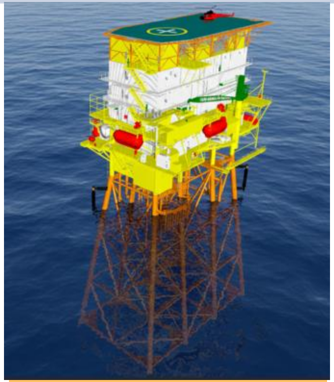
Integrity Level

Good	28
Fair	10
Near Poor	0
Poor	1
OOS	0
Total	**39**

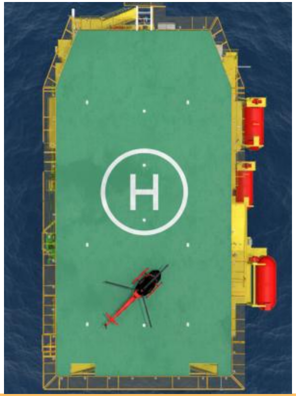




11.- Ek-A (Accomodation) *Type of Structure:* Octapod / *Year of Installation:* 2018 / *Number of Wells:* None



Side view



Bird's eye view

NDT results for each item

	Oily Drainage (atm pressure)	Diesel	Utility Air	Potable water for services	Raw water	Fire water	Instrument Air	Subtotal
Piping circuit	5	12	6	1	7	10	13	**54**
Diesel Tank	0	4	0	0	0	0	0	**4**
Diesel Filter	0	1	0	0	0	0	0	**1**
Air Filter	0	0	2	0	0	0	0	**2**
Air Receiver	0	0	1	0	0	0	2	**3**
Water Filter	0	0	0	0	2	0	0	**2**
Level Gauge	0	0	0	0	1	0	0	**1**
Water Tank	0	0	0	0	1	0	0	**1**

NDT results for each item **Total 68**

SC			2			1	7	**10**
RSC								**0**
RNC						2	4	**6**
NI	5	15	7	1	11	7		**46**
CID							3	**3**
NRLt							1	**1**
NC		2						**2**
OOS								**0**

NDT Summary

Standard Compliant	10
Repaired Standard-Compliant	0
Repaired Non-Compliant	6
Not inspected	46
Corrosion-Induced Degradation	3
Near to Retirement Limit thickness	1
Non Compliant	2
OOS	0

Integrity Level

Good	16
Fair	46
Near Poor	4
Poor	2
OOS	0
Total	**68**